Geometric Sequences: Recursive and Explicit Formulas

Easy level · 40 problems · A worked example comes before each part.

Name: _____ Date: _____

Key Idea

A **geometric sequence** multiplies by the same number every time. That number is the **common ratio** r.

$$r = \frac{\text{any term}}{\text{the term before it}} = \frac{a_2}{a_1}$$

Formula	Looks like	What it tells you
Recursive	$a_1 =$ first term, $\quad a_n = r \cdot a_{n-1}$	how to get the *next* term from the one before
Explicit	$a_n = a_1 \cdot r^{\,n-1}$	how to jump straight to *any* term a_n

Common Mistake

Divide to find r, do not subtract: for $3, 6, 12, \ldots$ $r = 6 \div 3 = 2$ (not $6 - 3 = 3$).
The exponent is $n - 1$, not n: $a_1 = a_1 \cdot r^0$. And a recursive formula **must** state a_1.

Part 1 Write the Recursive Formula

Problems 1–20

Watch Me — Worked Example

Write a recursive formula for $5,\ 15,\ 45,\ 135,\ \ldots$

Step 1 Find r: divide a term by the one before it $\quad r = 15 \div 5 = 3$ (check: $45 \div 15 = 3$ ✓)

Step 2 Write the first term $\quad a_1 = 5$

Step 3 Each term is r times the one before $\quad a_n = 3 \cdot a_{n-1}$

Answer $\quad \boldsymbol{a_1 = 5,\quad a_n = 3 \cdot a_{n-1}}$

Negative ratio? For $2, -6, 18, \ldots$: $r = -6 \div 2 = -3$, so $a_1 = 2$, $a_n = (-3) \cdot a_{n-1}$ (the signs alternate).

1. $2,\ 6,\ 18,\ 54,\ \ldots$

$r =$ _____

$a_1 =$ _____ $\quad a_n =$ _____

2. $1,\ 2,\ 4,\ 8,\ \ldots$

$r =$ _____

$a_1 =$ _____ $\quad a_n =$ _____

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3. $5,\ 10,\ 20,\ 40,\ \ldots$

$r =$ _____

$a_1 =$ _____ $a_n =$ _____

4. $3,\ 12,\ 48,\ 192,\ \ldots$

$r =$ _____

$a_1 =$ _____ $a_n =$ _____

5. $4,\ 12,\ 36,\ 108,\ \ldots$

$r =$ _____

$a_1 =$ _____ $a_n =$ _____

6. $1,\ 5,\ 25,\ 125,\ \ldots$

$r =$ _____

$a_1 =$ _____ $a_n =$ _____

7. $6,\ 12,\ 24,\ 48,\ \ldots$

$r =$ _____

$a_1 =$ _____ $a_n =$ _____

8. $2,\ 20,\ 200,\ 2000,\ \ldots$

$r =$ _____

$a_1 =$ _____ $a_n =$ _____

9. $7,\ 14,\ 28,\ 56,\ \ldots$

$r =$ _____

$a_1 =$ _____ $a_n =$ _____

10. $3,\ 9,\ 27,\ 81,\ \ldots$

$r =$ _____

$a_1 =$ _____ $a_n =$ _____

11. $1,\ -2,\ 4,\ -8,\ \ldots$

$r =$ _____

$a_1 =$ _____ $a_n =$ _____

12. $2,\ -6,\ 18,\ -54,\ \ldots$

$r =$ _____

$a_1 =$ _____ $a_n =$ _____

13. $5,\ -10,\ 20,\ -40,\ \ldots$

$r =$ _____

$a_1 =$ _____ $a_n =$ _____

14. $64,\ 32,\ 16,\ 8,\ \ldots$

$r =$ _____

$a_1 =$ _____ $a_n =$ _____

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15. $81,\ 27,\ 9,\ 3,\ \ldots$

$r =$ _____

$a_1 =$ _____ $a_n =$ _____

16. $-1,\ -2,\ -4,\ -8,\ \ldots$

$r =$ _____

$a_1 =$ _____ $a_n =$ _____

17. $-3,\ -9,\ -27,\ -81,\ \ldots$

$r =$ _____

$a_1 =$ _____ $a_n =$ _____

18. $10,\ -10,\ 10,\ -10,\ \ldots$

$r =$ _____

$a_1 =$ _____ $a_n =$ _____

19. $4,\ 20,\ 100,\ 500,\ \ldots$

$r =$ _____

$a_1 =$ _____ $a_n =$ _____

20. $1000,\ 100,\ 10,\ 1,\ \ldots$

$r =$ _____

$a_1 =$ _____ $a_n =$ _____

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Part 2 Write the Explicit Formula — Problems 21–40

Watch Me — Worked Example

Write an explicit formula for $4,\ 8,\ 16,\ 32,\ \ldots$

Step 1 Find a_1 and r $\quad a_1 = 4, \quad r = 8 \div 4 = 2$

Step 2 Substitute into $a_n = a_1 \cdot r^{\,n-1}$ $\quad \boldsymbol{a_n = 4 \cdot 2^{\,n-1}}$

Check $n = 3$ $\quad a_3 = 4 \cdot 2^2 = 16$ ✓

Fraction or negative ratio? Put it in parentheses: $81, 27, 9, \ldots$ gives $a_n = 81 \cdot \left(\frac{1}{3}\right)^{n-1}$; $1, -2, 4, \ldots$ gives $a_n = (-2)^{n-1}$.

21. $3,\ 6,\ 12,\ 24,\ \ldots$

$a_1 =$ _____ $\quad r =$ _____

$a_n =$ _____

22. $1,\ 3,\ 9,\ 27,\ \ldots$

$a_1 =$ _____ $\quad r =$ _____

$a_n =$ _____

23. $2,\ 10,\ 50,\ 250,\ \ldots$

$a_1 =$ _____ $\quad r =$ _____

$a_n =$ _____

24. $4,\ 8,\ 16,\ 32,\ \ldots$

$a_1 =$ _____ $\quad r =$ _____

$a_n =$ _____

25. $5,\ 15,\ 45,\ 135,\ \ldots$

$a_1 =$ _____ $\quad r =$ _____

$a_n =$ _____

26. $1,\ 4,\ 16,\ 64,\ \ldots$

$a_1 =$ _____ $\quad r =$ _____

$a_n =$ _____

27. $2,\ 4,\ 8,\ 16,\ \ldots$

$a_1 =$ _____ $\quad r =$ _____

$a_n =$ _____

28. $6,\ 18,\ 54,\ 162,\ \ldots$

$a_1 =$ _____ $\quad r =$ _____

$a_n =$ _____

29. $1,\ 10,\ 100,\ 1000,\ \ldots$

$a_1 =$ _____ $\quad r =$ _____

$a_n =$ _____

30. $3,\ 15,\ 75,\ 375,\ \ldots$

$a_1 =$ _____ $\quad r =$ _____

$a_n =$ _____

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31. $2,\ -4,\ 8,\ -16,\ \ldots$

$a_1 =$ _____ $\quad r =$ _____

$a_n =$ _____

32. $1,\ -3,\ 9,\ -27,\ \ldots$

$a_1 =$ _____ $\quad r =$ _____

$a_n =$ _____

33. $4,\ -8,\ 16,\ -32,\ \ldots$

$a_1 =$ _____ $\quad r =$ _____

$a_n =$ _____

34. $32,\ 16,\ 8,\ 4,\ \ldots$

$a_1 =$ _____ $\quad r =$ _____

$a_n =$ _____

35. $27,\ 9,\ 3,\ 1,\ \ldots$

$a_1 =$ _____ $\quad r =$ _____

$a_n =$ _____

36. $-2,\ -6,\ -18,\ -54,\ \ldots$

$a_1 =$ _____ $\quad r =$ _____

$a_n =$ _____

37. $-1,\ -2,\ -4,\ -8,\ \ldots$

$a_1 =$ _____ $\quad r =$ _____

$a_n =$ _____

38. $5,\ -5,\ 5,\ -5,\ \ldots$

$a_1 =$ _____ $\quad r =$ _____

$a_n =$ _____

39. $7,\ 21,\ 63,\ 189,\ \ldots$

$a_1 =$ _____ $\quad r =$ _____

$a_n =$ _____

40. $5000,\ 500,\ 50,\ 5,\ \ldots$

$a_1 =$ _____ $\quad r =$ _____

$a_n =$ _____

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